Model choice - comparison

AI Invoice Assistant - 3 October 2026 - based on published benchmarks

Model (Ollama)	Size	Fits 4 GB GPU	Instruction following (IFEval, higher = better)	Tool / format calling (BFCL, higher = better)	Licence	Verdict
`llama3.2:1b`	1.3 GB	yes	59.5	25.7	Llama 3.2 Community	Too weak
`qwen2.5:1.5b`	1.0 GB	yes	42.5*	n/p	Apache 2.0	Too weak
`llama3.2:3b`	2.0 GB	yes	77.4	67.0	Llama 3.2 Community	**Choose** - given by the task
`qwen2.5:3b`	1.9 GB	yes	58.2*	n/p	Qwen Research: no commercial use	Excluded (licence)
`phi4-mini`	2.5 GB	yes	n/p	n/p	MIT	Possible, little data
`qwen3:4b`	2.5 GB	yes	81.2*	57.6 (v3)	Apache 2.0	**Fallback no. 1**
`gemma3:4b`	3.3 GB	tight	90.2	n/p	Gemma Terms	Fallback no. 2 (slower)
`qwen2.5:7b`	4.7 GB	no	71.2*	n/p	Apache 2.0	Too slow here
`llama3.1:8b`	4.9 GB	no	80.4	67.1	Llama 3.1 Community	Too slow here

n/p = not published by the vendor. * = Qwen reports the stricter "strict prompt" IFEval variant, so its numbers read a few points low against the others.

Recommendation

- **Keep llama3.2:3b.** It is the model given in the task, fits the GPU, and has the best published format-calling score (BFCL 67.0) of all models that fit. Its weaker instruction following is covered by the design: Python validates every command, asks for confirmation and does all calculations.
- **If it is not good enough in practice, switch to qwen3:4b** (thinking mode off): better instruction following, Apache 2.0, same disk size class. Only the model line in the Modelfiles changes.
- **gemma3:4b** has the highest instruction-following score but only just fits the 4 GB GPU, so answers will be slower.

Why the others are out

- **1B / 1.5B models:** too weak at following instructions for multi-step conversations.
- **qwen2.5:3b:** its licence does not allow commercial use, and company X is a business.
- **7B / 8B models:** do not fit the 4 GB GPU, so most of the work runs on the CPU and replies become much slower, for a small quality gain.

Read with care: vendor-reported scores measured with different settings (e.g. Llama 3.1 8B: IFEval 80.4 in Meta's card, 75.9 in Qwen's comparison). Treat differences under about 5 points as a tie. Benchmarks measure general skills, not our exact conversations; the bots must still be checked against the scenarios in requirements.pdf.

Sources: huggingface.co/meta-llama/Llama-3.2-3B-Instruct (Llama 3.2 and 3.1) - qwenlm.github.io/blog/qwen2.5-llm (Qwen2.5, licences) - huggingface.co/Qwen/Qwen3-4B-Instruct-2507 (Qwen3-4B non-thinking) - ai.google.dev/gemma/docs/core/model_card_3 (Gemma 3 IT) - huggingface.co/microsoft/Phi-4-mini-instruct - sizes: Ollama library, 4-bit versions installed here.